

Segmenting Minerals in Rotational Lunar Thin Section Images

The Moon is my beautiful impossibility.

Ping Wang

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This guide introduces how to turn a rotation series of thin section photomicrographs into a mineral map by using the Phase Segmentation app [1]. You will identify the four major rock-forming minerals: opaques, plagioclase, pyroxene, and olivine, and glass in a lunar rock, separate them by phases, and export a short animation of each mineral.

1 Basic Concepts

1.1 Thin sections

A thin section is a slice of rock ground down to about 30 micrometres (0.03 mm) and glued to a glass slide. At that thickness most rock-forming minerals become transparent, so light shines through them and we can study how each mineral bends and polarizes that light. The 30 μm standard matters because the colors you see under crossed polars are calibrated to it.

1.2 PPL & XPL

The polarizing microscope has two polarizing filters, and you switch between two setups:

Plane-polarized light (PPL) uses only the lower polarizer. Here you examine a mineral by its natural color, its relief (how strongly its outline stands out, which reflects its refractive index), its cleavage (planes of weakness that show as straight cracks), and its shape.

Cross-polarized light (XPL) adds a second polarizer rotated 90° to the first (hence "crossed"). Now you see interference colors produced by the mineral splitting light into two rays that travel at different speeds. The strength of this effect is called *birefringence*, and different minerals show characteristically different interference colours compared against the Michel-Lévy color chart.

1.3 Extinction: Why You Need to Use Rotational Series

When you rotate the stage under crossed polars, an ordinary crystalline (anisotropic) mineral goes dark four times per full turn, once every 90° , and shows its bright interference color in between. Some materials, though, stay dark at every angle:

Isotropic materials such as volcanic glass and cubic minerals never produce interference colors, so they are always dark in XPL — but they still transmit light, so they look clear and bright in PPL.

Opaque minerals, metal oxides such as ilmenite, are dark in both PPL and XPL because no light passes through them at all.

Here is the catch that makes a single PPL/XPL image insufficient: a normal mineral that happens to be at extinction in one frame also looks dark in that frame. You cannot tell it apart from truly always-dark material from one image.

By photographing the section at many rotation angles, you can see which grains are dark at every angle (opaque or glass) and which are only momentarily dark (crystalline minerals at extinction). That is exactly why the input is a 36-frame rotational series, and it is the principle the app's automatic steps rely on.

2 Lunar Rock and Lunar Minerals

The workflow is built for lunar rock thin sections. The phases the Phase Segmentation app separates are: opaque oxides, glass, plagioclase, pyroxene, and olivine. You adjust two thresholds to segment opaques and glass. Then you will identify the three crystalline silicates.

Table 1: Key optical properties of common lunar silicate minerals in thin sections.

Property	Plagioclase	Pyroxene	Olivine
Color in PPL	Colorless	Pale brown	Colorless
Relief	Low (barely stands out)	Moderate to high	High (strong outline)
Cleavage / fractures	Twin lamellae	Two cleavages intersecting at approximately 90°	No cleavage; irregular cracks
XPL interference colors	Low: first-order grey and white	Moderate: first- to second-order colors	High: vivid second- to third-order pink, green and blue
Diagnostic feature	Polysynthetic ("striped") twinning	Two nearly right-angle cleavages with inclined extinction	Very high birefringence

A quick way to remember the three: Plagioclase is the low-relief, dull-grey, stripey one; olivine is the high-relief, brilliantly colored one; pyroxene sits in between and shows two cleavages at nearly a right angle. The stripes in plagioclase (its twinning) are the most reliable single feature: watch the light and dark lamellae wink to extinction at slightly different stage positions.

3 Using the App, Tab by Tab

You work left to right through the tabs. Each step uses the results of the one before it, so do them in order.

Tab 0: Load and FOV. Upload your zip file of PPL/XPL frames. The app finds the circular edge of the view and shows it: yellow is the detected circle, green is where it will crop, red is the center. If the green circle sits slightly inside the bright area, that is intended: it trims the dark rim. You nudge the sliders if the circle is clearly off.

Tab 1: Drift Check. As the stage rotated, its centre of rotation may have drifted a few pixels. This tab measures that. The plot's "smoothed creep" line is the slow, one-directional wander across the series, the small wobble you would otherwise notice near the end of looping animation. If it exceeds about one pixel, the app recommends correcting it. Note the rotation total too: a series that reads about 0° is a non-rotating clip, which the app handles by passing it through unchanged.

Tab 2-3: Crop and Stabilize. This crops every frame to the field of view, makes the outside transparent, and applies the drift correction while keeping the intended rotation. Leave the mode on "adaptive": it only applies the correction if a self-check confirms it actually steadies the series, so it can never make things worse. The gallery shows a few registered frames.

Tabs 4-5: Label Opaques and glass. Using the rotation series, the app marks the always-dark

material for you. Opaque grains are dark in XPL at every angle and dark in PPL; glass is dark in XPL at every angle but bright in PPL. Adjust the two threshold sliders until the colored overlay matches what you see. These regions are then excluded from your manual labelling, because they are already identified.

Tab 6: Label Crystalline Silicates.

1. Choose whether to label on the PPL or the XPL image. XPL is usually easier for telling the minerals apart by their interference colors and twinning; PPL is better for relief and cleavage. You can rebuild on either at any time.
2. Click **Build superpixels**. The app divides the image into a few hundred small regions, called "superpixels", that follow grain boundaries, so one click selects a whole coherent patch rather than a single pixel. Raise the zoom slider to make clicking easier.
3. Pick a phase (plagioclase, pyroxene, or olivine), then click the superpixels that belong to it. Click again to remove a label. Work one mineral at a time; the status line keeps a running count.
4. Use the identification workflow below to decide what each grain is. Do not guess from a single frame, but where possible, look at how a grain behaves across the rotation.
5. Auto-fill by color is an optional shortcut: it assigns every unlabelled superpixel to whichever labelled mineral it most resembles in color. Treat its result as a rough first pass and correct the mistakes; it does not understand relief, cleavage, or twinning the way you do.

Tab 7: Export the phase GIFs. Each mineral is exported as its own animation on a transparent background, plus a combined map. Keep the default "stabilized" style: the grain is held still while its appearance cycles in XPL you watch it move through extinction in place. Choose the XPL source to get that colour-cycling effect. The alternative "rotating" style spins each phase with the original series instead. Download the GIFs and the zip.

4 Walking Through Tab 6

When you reach a grain in Tab 6, work through it like this:

First, is it dark in XPL at every rotation? If so, it is opaque or glass and is already auto-labelled; so skip it. If the auto step missed it, dark-in-PPL means opaque, bright-in-PPL means glass.

If it does show color in XPL, it is one of your three crystalline minerals. Now combine clues rather than relying on one:

- Check relief in PPL. Low relief points to plagioclase. Moderate-to-high relief points to pyroxene or olivine.
- Check the XPL interference colours. Dull 1st-order grey and white suggest plagioclase; moderate colours suggest pyroxene; brilliant, high-order colors suggest olivine.
- Look for the diagnostic feature. Parallel light/dark stripes (twinning) confirm plagioclase. Two cleavage traces meeting at about 90° confirm pyroxene. No cleavage plus vivid colours and cracked, sometimes reddish grains confirm olivine.
- Use the rotation. Watch a grain across several frames: plagioclase twins extinguish in alternating bands; pyroxene and olivine simply cycle to extinction and back.

When two minerals seem plausible, let the diagnostic feature, twinning versus right-angle cleavage, break the tie.

5 Exports

Your default exports are doubly stabilized: the frames have the stage drift removed, and each phase is held fixed in place rather than spinning. So an olivine GIF, for example, shows an olivine grain sitting still while its interference color brightens and darkens through extinction, the same effect you would see by rotating the stage yourself while staring at one crystal. That "color cycling in place" is the signature of an anisotropic mineral, and it is a good visual check that you labelled a real crystalline grain and not a patch of glass which would stay dark or an opaque grain which would stay black.

Glossary

Anisotropic: a mineral that affects polarized light and therefore shows interference colours and extinction under crossed polars.

Birefringence: the difference in speed between the two rays a mineral splits light into; higher birefringence means brighter, higher-order interference colors.

Cleavage: planar weaknesses in a mineral that appear as sets of straight parallel cracks; their number and angle help identify the mineral.

Extinction: the position, every 90° of stage rotation, at which an anisotropic mineral appears dark under crossed polars.

Isotropic: a material, such as glass, that does not affect polarized light and stays dark under crossed polars at all angles.

Interference colors: the colours seen under crossed polars, produced by birefringence and compared against the Michel-Lévy chart.

PPL / XPL: plane-polarized light (one polarizer) and cross-polarized light (two crossed polarizers).

Relief: how strongly a grain's outline stands out in PPL, reflecting the difference between its refractive index and that of the mounting medium.

Superpixel: a small cluster of neighbouring pixels grouped by the app so that one click labels a whole coherent region.

Polysynthetic Twinning: the parallel light / dark lamellae characteristic of plagioclase, which extinguish in alternating bands as the stage rotates.

References

- [1] "Phase Segmentation," *Hugging Face Spaces*, Hugging Face. [Online]. Available: <https://huggingface.co/spaces/pingwang2026/phase-segmentation>